GO-IAQG

GLOBAL OPEN INDOOR AIR QUALITY GOVERNANCE

From scores and predictions to examinable evidence, established authority, admissible execution and verified outcomes.

Expanded global founding technical brief | October 2026

Prepared by TA-14 Authority Governance Institution LLC

For governments, public-health institutions, school systems, engineers, building owners, community organizations and potential demonstration partners.

STATUS: Proposed educational and technical governance framework. This document is not a certification, regulatory mandate, completed independent validation or proof of health outcomes. Illustrative images and any displayed numerical examples are not real pilot data.





Executive proposition

The world does not need to discard air-quality indices. It needs a defensible way to determine when observations and interpretations are sufficient to justify a specific action, who may authorize that action, and how the result will be verified. GO-IAQG proposes that missing layer of governance.

A reading is an observation. A score is a transformation. A prediction is an inference. None is automatically an instruction, a legal authorization, a verified intervention, or a demonstrated health outcome.

GO-IAQG is being developed by TA-14 Authority Governance Institution LLC as an open educational and governance proposition. Its methods require evaluation in real buildings and against applicable laws, standards and independent technical scrutiny.

The complete architecture, not merely an execution gate

Layer	Purpose and boundary
Atmospheric Integrity Records (AIR)	Preserve raw observations, provenance, context, continuity, adjustments and limitations.
Admissible Computation Architecture (ACA)	Govern which evidence and transformations may be used in an interpretation; keep inference distinct from observation.
Admissible Standing Architecture (ASA)	Establish actor, capacity, standing basis and authority relationship.
Admissible Federation Architecture (AFA)	Share evidence and governance context across organizations without exporting execution authority.
Execution Authority Boundary Architecture (EABA)	Separate a recommendation or instruction from locally established execution permission.
Admissible Execution Architecture (AEA)	Examine proposed consequences before commit and execution; retain decisions and receipts.
Outcome examination	Assess what happened and whether intended results were supported by evidence.

The failure of score-only governance

Scores, color bands and predictive dashboards have legitimate uses: screening, public communication, prioritization and trend recognition. Their failure mode occurs when a simplified representation is treated as sufficient proof for a decision it was not designed to support.

What a score can do	What it cannot establish by itself
Summarize selected measured or modeled inputs	That all relevant pollutants, rooms, occupants and time periods were covered
Communicate an estimated category or risk	That sensor accuracy, calibration and data completeness are adequate
Flag a potential problem	Which source caused the condition or which intervention will work
Track an aggregate trend	That an individual child experienced a particular exposure or health effect
Support triage and prioritization	Who is authorized to act and whether action occurred
Show a lower value after intervention	That the intervention caused the change or that protection persists

This is not a claim that all indices or models are flawed. It is a claim about the limits of inference and authorization: the validity of any conclusion depends on the decision being made, its evidentiary requirements and the scope of the underlying observations.

The governing question

Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

Technical evidence requirements

A minimum Atmospheric Integrity Record should support inspection of the following categories, with requirements scaled to the intended decision and the available equipment.

Category	Example record fields
Identity and origin	Record ID; instrument and firmware ID; responsible organization; room/zone; collection method
Time and continuity	Timestamp and time zone; frequency; coverage; gaps; clock synchronization; retention
Measurement integrity	Pollutant or parameter; units; method; calibration and QA status; detection limits; uncertainty
Operational context	Occupancy; HVAC mode; outdoor conditions; filter and maintenance status; unusual events
Transformation lineage	Original raw value; corrections; exclusions; adjusted value; model version; computation inputs
Decision lineage	Question examined; reviewer; authority basis; disposition; rationale; decision time
Execution and outcome	Approved work; implementer; change log; execution receipt; post-action evidence; residual uncertainty

Raw and adjusted records are not interchangeable

Raw observations should remain available as the primary record. Corrected, normalized, modeled or imputed values must be stored separately with explicit transformation lineage. A secure ledger can preserve a record of what was received, but it cannot retroactively make an inaccurate sensor accurate.

Admissibility is purpose- and jurisdiction-dependent

The word “Admissible” here describes the proposed TA14 governance examination. It does not claim automatic legal admissibility in any court, procurement, administrative proceeding or national regulatory framework.

Decision authority and fail-closed behavior

An environmental recommendation may be scientifically plausible yet unauthorized for a specific building. GO-IAQG therefore separates evidence quality, standing, delegated power, safety constraints and execution.

Disposition	Illustrative use
ALLOW	Evidence, standing and applicable authority meet the defined threshold for the proposed action.
HOLD	Evidence is missing, stale, incomparable or insufficient; request further observation or review.
DENY	The requested action conflicts with governing rules or the actor lacks authority to perform it.
ESCALATE	A competent higher authority or specialized professional must determine the next step.

These dispositions are illustrative. In urgent hazards, the framework must not delay legally required emergency responses. Applicable emergency plans and competent human authorities remain controlling.

Federation without authority transfer

A school in one country may learn from a comparable school elsewhere; a health agency may share a measurement protocol; researchers may publish anonymized evidence. But shared methods, templates or AI-generated recommendations do not create the local right to order building work, direct occupants or impose sanctions.

Illustrative pilot: one school, one bounded decision

Scenario: a school reports elevated occupied-hour CO2 and concerns about ventilation. The example is hypothetical; no displayed values or outcomes should be read as real-world evidence.

Stage	Evidence and action
1. Scope	Identify rooms, school authority, occupant groups, objectives, applicable requirements and who can approve interventions.
2. Baseline	Collect time-stamped CO2, occupancy, outdoor conditions, HVAC schedules, sensor QA and relevant operational records.
3. Interpretation	Examine whether the pattern supports a ventilation investigation; avoid equating CO2 directly with all pollutants or individual health outcomes.
4. Standing and authority	Confirm who can approve inspections, operational adjustments, procurement and communication to families.
5. Boundary decision	ALLOW a bounded authorized diagnostic step, or HOLD/ESCALATE when evidence or authority is insufficient.
6. Execution	Record the actual HVAC inspection, adjustment or repair, who performed it, when and with what authorization.
7. Verification	Compare comparable occupied periods, document uncertainty and competing explanations, and monitor persistence.
8. Publication	Publish a privacy-conscious account of observations, actions, limitations and remaining work.

What success would actually mean

Success is not “the dashboard turned green.” It is an auditable chain showing that an authorized action occurred, that relevant environmental conditions changed under reasonably comparable conditions, and that uncertainty and residual risks were not concealed. Demonstrating reduced disease, improved cognition or attendance would require additional study designs and evidence.

Implementation blueprint for institutional partners

Phase A | Governance scoping

Map local standards, decision-makers, legal authority, facility constraints, stakeholder rights, procurement, privacy and record-retention duties. Define the precise proposed consequences that the gate will examine.

Phase B | Evidence design

Choose relevant pollutants and indicators; define QA thresholds, minimum sampling duration, contextual variables, coverage rules, missing-data handling, transformations and reviewer access.

Phase C | Bounded pilot

Use a controlled scope with written approval, baseline records, a defined intervention class, named reviewers, explicit HOLD conditions and an outcome evaluation plan established before implementation.

Phase D | Independent examination

Invite qualified third-party review of provenance, assumptions, authority chain, disposition, execution receipts and outcome claims. Publish findings including negative and inconclusive results.

Phase E | Replication and federation

Share validated methods and anonymized examples where permitted; require each new jurisdiction and facility to establish its own applicable authority and local implementation plan.

Suggested pilot performance measures

Domain	Example measure
Evidence completeness	Percent of required fields present; gap duration; calibration or QA exceptions
Governance integrity	Percent of decisions with recorded authority, rationale and disposition
Execution traceability	Percent of actions linked to approvals, work records and completion evidence
Environmental response	Pollutant- or parameter-specific changes with contextual comparability and uncertainty
Continuity	Persistence of outcomes over prespecified follow-up periods
Equity and trust	Coverage across rooms and occupant groups; timely, understandable communication

Safeguards, limitations and non-claims

• GO-IAQG does not replace qualified engineers, health professionals, established standards, regulators or legal processes.

• A score is not inherently invalid; its interpretation must remain bounded by its underlying data and purpose.

• A monitoring result cannot by itself establish individual dose, causation, disease prevention or clinical benefit.

• Cryptographic integrity, signatures and logs support provenance; they do not guarantee sensor validity or correct interpretation.

• Children and vulnerable populations warrant appropriate protections, but data collection and disclosure must respect privacy and local law.

• AI can support computation and recommendations, but capability is not authority. A model output is not permission to execute.

• No global adoption, regulatory endorsement, legal admissibility or verified health benefit is asserted by this educational brief.

Invitation to examine, not an assertion of endorsement

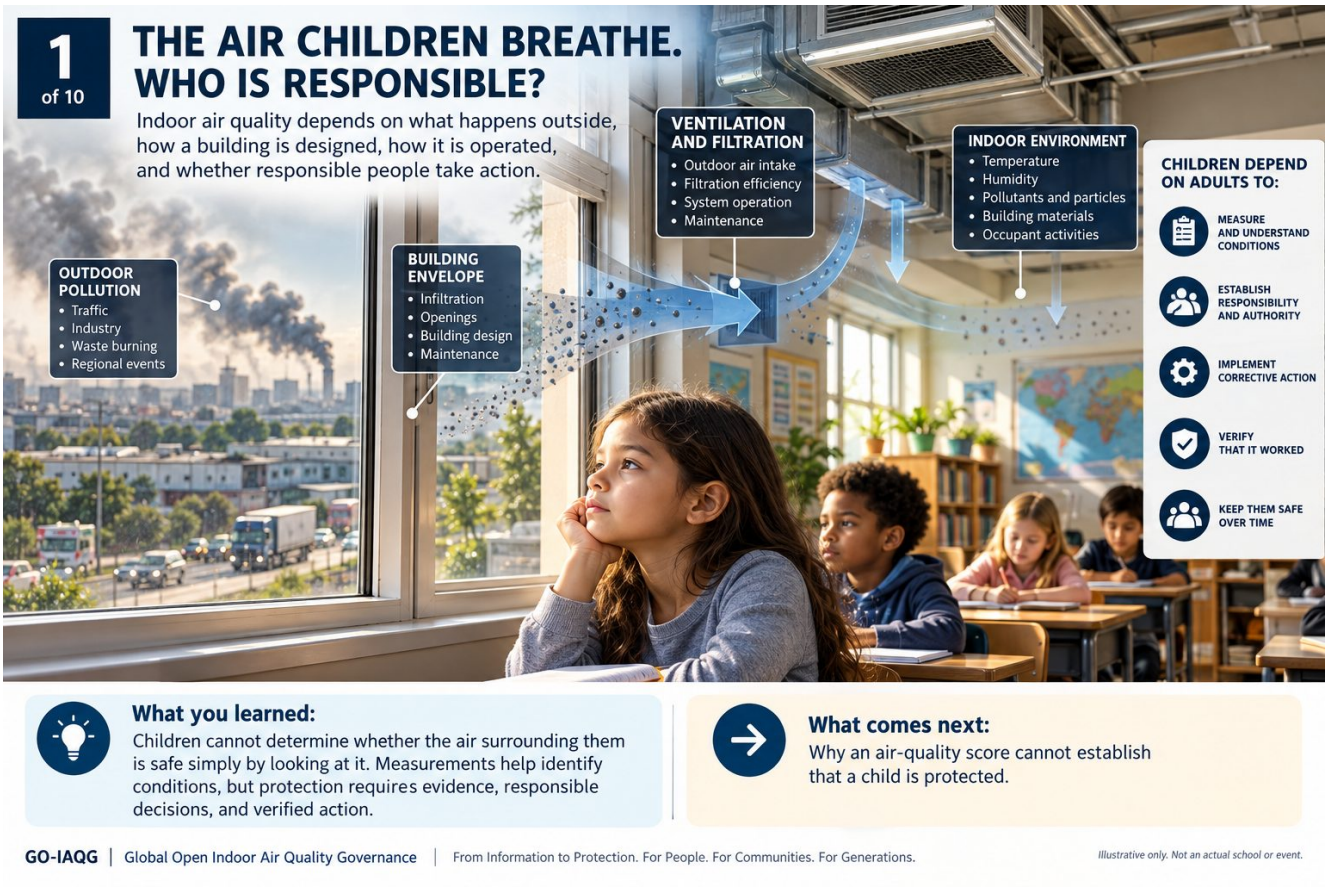
TA14 welcomes written technical dialogue with institutions, building owners, school systems, researchers, public-health authorities and solution providers about bounded evaluations. Initial commercial exploration or introductory demonstration does not require a fee or registration; any formal examination or deployment would need a separately agreed scope and terms.

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No admissible evidence. No admissible execution.

01 / 10 The air children breathe: who is responsible?

Indoor conditions arise from the interaction of outdoor sources, building envelopes, mechanical systems, indoor activities and human decisions.



What the image teaches

An acceptable-looking classroom is not proof of acceptable exposure. A responsible assessment defines occupants, spaces, pollutants of concern, observation periods, and the person authorized to decide what happens next.

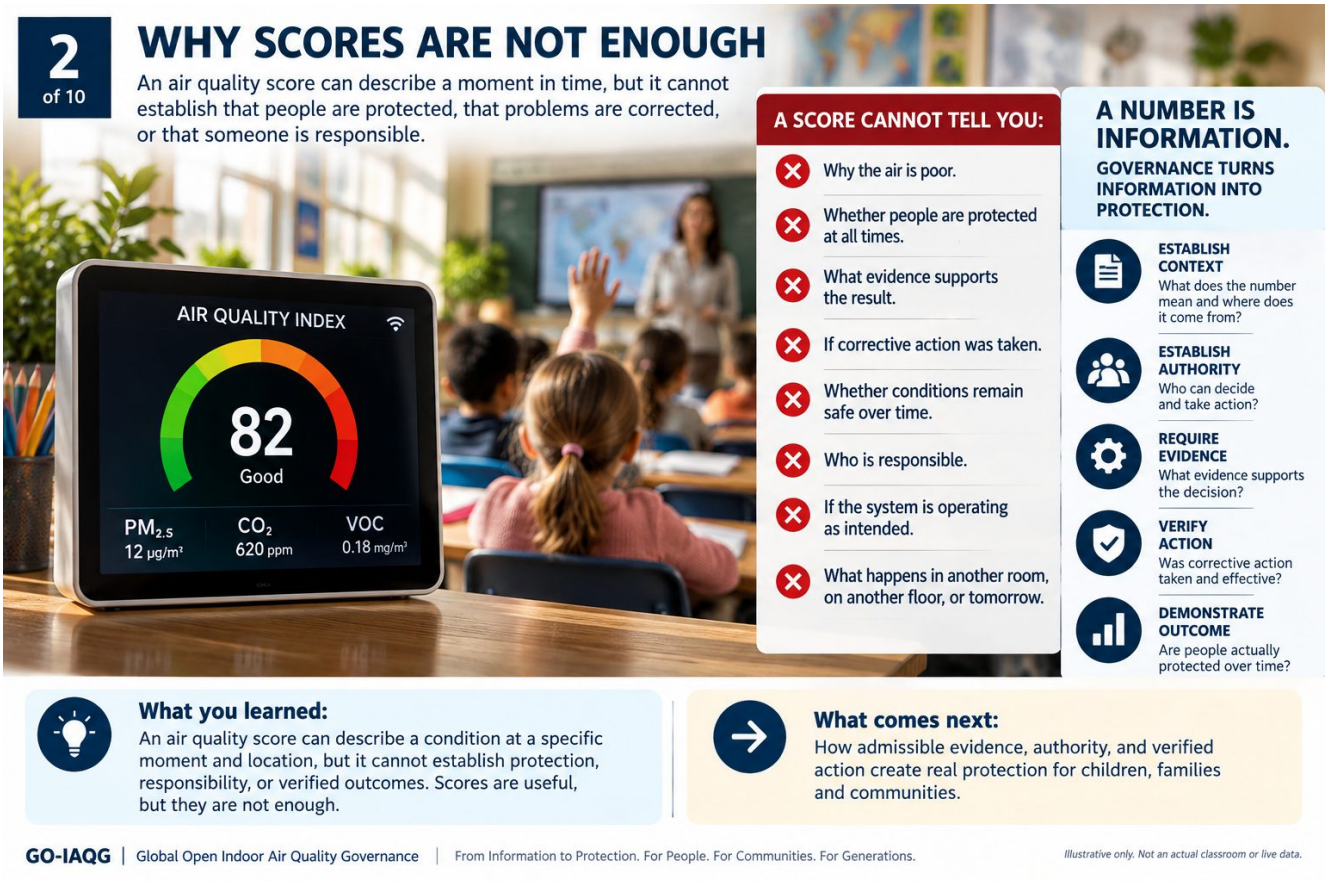
Examination question

Where and when was the condition observed? Which children or occupants could be affected? What decision is actually being proposed?

Illustrative educational artwork only. Depicted persons, locations, charts, readings and outcomes are not documentary evidence.

02 / 10 Why scores are not enough

An index is a communication layer, not a complete chain of custody, a causal explanation, or an authorization to intervene.



What the image teaches

Retain pollutant-specific readings, units, methods, time aggregation, coverage, missing data, uncertainty and model assumptions. A favorable index must not conceal a pollutant-specific concern or a short high-exposure event.

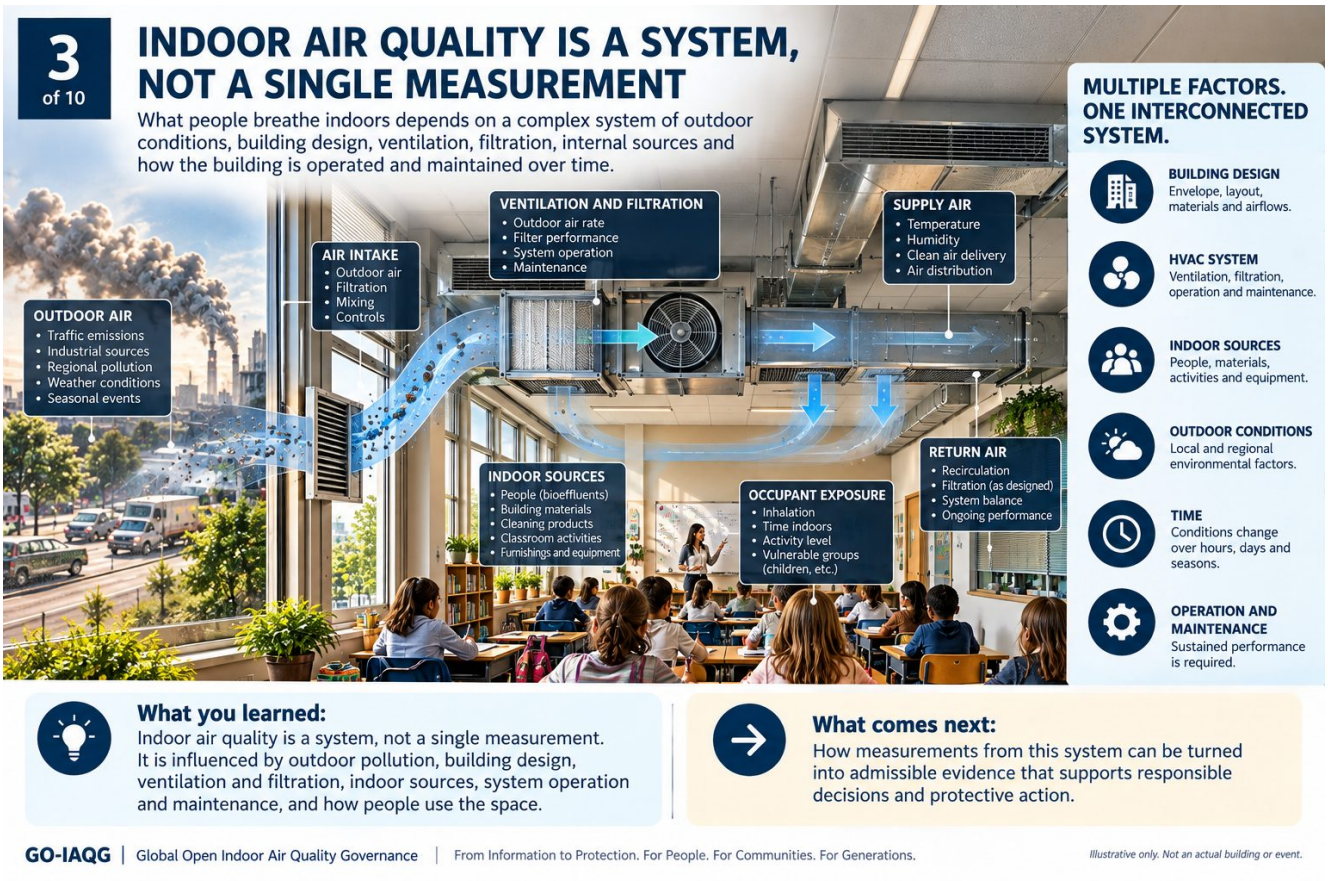
Examination question

Can an independent examiner reconstruct the score, inspect the source data, and identify what the score cannot establish?

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03 / 10 Indoor air quality is a system

Ventilation rates, filtration, infiltration, source strength, occupancy, maintenance and meteorology can change the interpretation of the same sensor reading.



What the image teaches

Consider sensor placement, occupied and unoccupied periods, outdoor conditions, system operating mode, cleaning events, construction work and plausible alternative explanations. Not every relevant factor can be measured continuously; record those limitations.

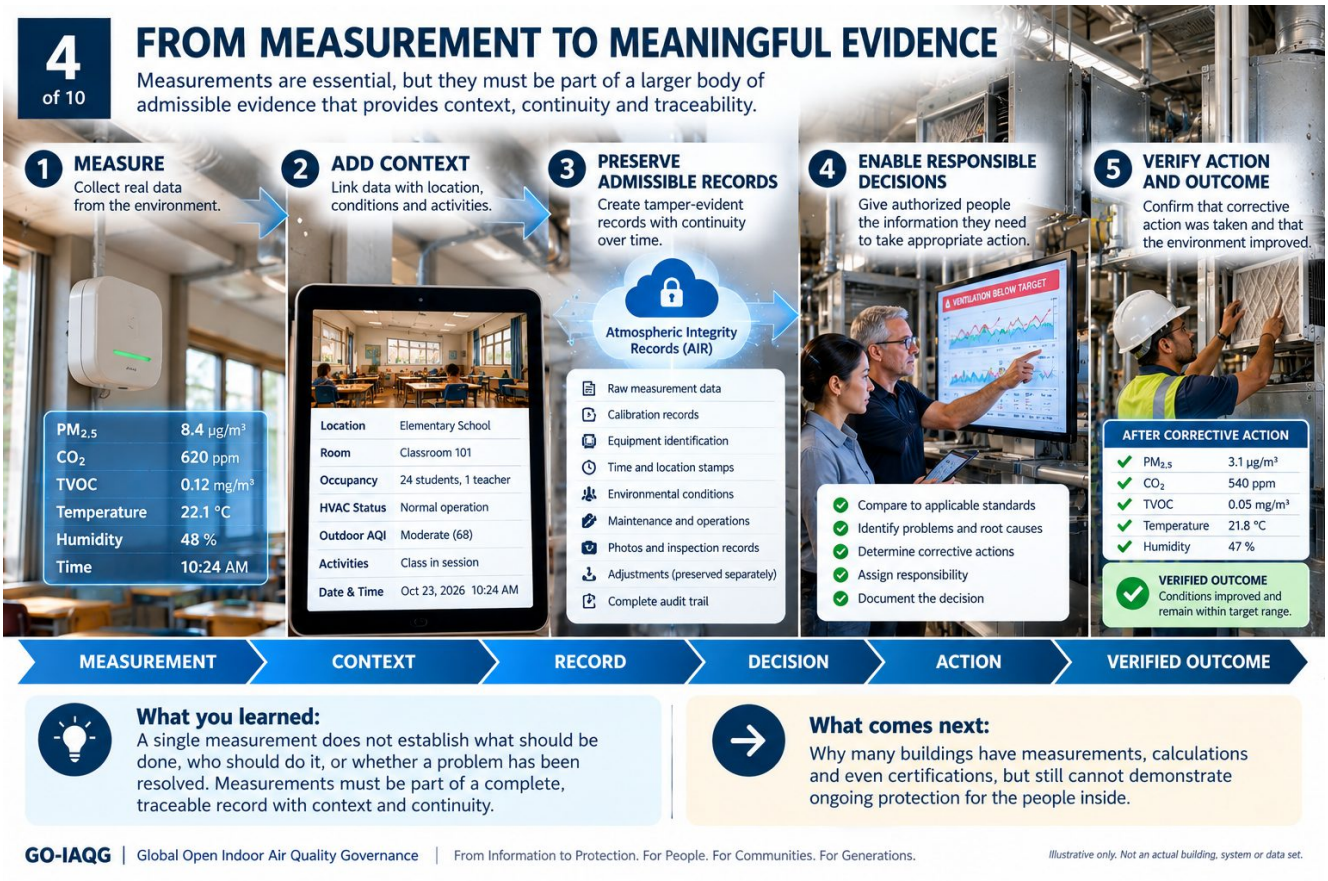
Examination question

Were the variables relevant to the intended decision observed or explicitly identified as unknown?

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04 / 10 From measurement to meaningful evidence

Atmospheric Integrity Records (AIR) are intended to preserve raw observations and the provenance necessary to evaluate their meaning.



What the image teaches

Preserve original and adjusted values separately; include instrument identity, calibration or quality status, timestamp basis, location, units, sampling frequency, gaps, corrections, transformations, source records and reviewer decisions. Tamper evidence is useful but does not by itself establish sensor accuracy.

Examination question

Can a qualified reviewer reconstruct a conclusion without relying on a dashboard screenshot or undocumented calculation?

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05 / 10 Establishing clear responsibility

A recommendation is not automatically an authorized instruction. Authority, capacity and standing must be established for the specific consequence.



What the image teaches

The Admissible Standing Architecture distinguishes ACTOR, CAPACITY, STANDING BASIS and AUTHORITY RELATIONSHIP. The owner, school authority, facility operator, contractor, regulator and reviewer may have different powers and duties. Record jurisdiction, delegation, decision scope and escalation.

Examination question

Who can authorize the action now, what is the basis for that authority, and where does it stop?

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06 / 10 From observation to protection

The TA14 chain separates what was observed from what may be executed: Reality, Record, Continuity, Admissibility, Binding, Commit, Execution, Outcome.



What the image teaches

At the boundary, the disposition may be ALLOW, HOLD, DENY or ESCALATE. A missing calibration record may require HOLD; absent authority may require DENY or ESCALATE. Emergency procedures and legally required duties remain governed by applicable local rules.

Examination question

Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

Illustrative educational artwork only. Depicted persons, locations, charts, readings and outcomes are not documentary evidence.

07 / 10 Real protection in real places

Schools, hospitals, homes and public facilities have different populations, exposures, operational limits and decision-makers.



What the image teaches

Schools may need occupied-hour evidence and parent communications; healthcare facilities may require specialized infection-control and engineering review; housing requires attention to landlord and tenant responsibilities; public buildings need transparent procurement and records.

Examination question

What evidence and authority are specific to this building, population and proposed action?

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08 / 10 Verified outcomes: not merely completed work

A completed work order proves an activity was documented; it does not prove the intended environmental result.



What the image teaches

A before/after comparison should preserve occupancy, outdoor conditions, system settings, measurement methods and uncertainty. Where feasible, use matched comparison periods or controls. Distinguish equipment function, concentration changes, exposure reduction and health outcomes; these are not interchangeable.

Examination question

What improved, for whom, for how long, and what competing explanations remain?

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09 / 10 Stronger communities through accountable air governance

Governance can make environmental claims more reviewable, but improvements in health, attendance, learning or productivity must be measured rather than assumed.



What the image teaches

A useful public record distinguishes observed environmental change, documented decisions, implementation quality, costs, unresolved problems and distributional effects. Communities should be able to understand what was measured, who decided, and whether promises were kept.

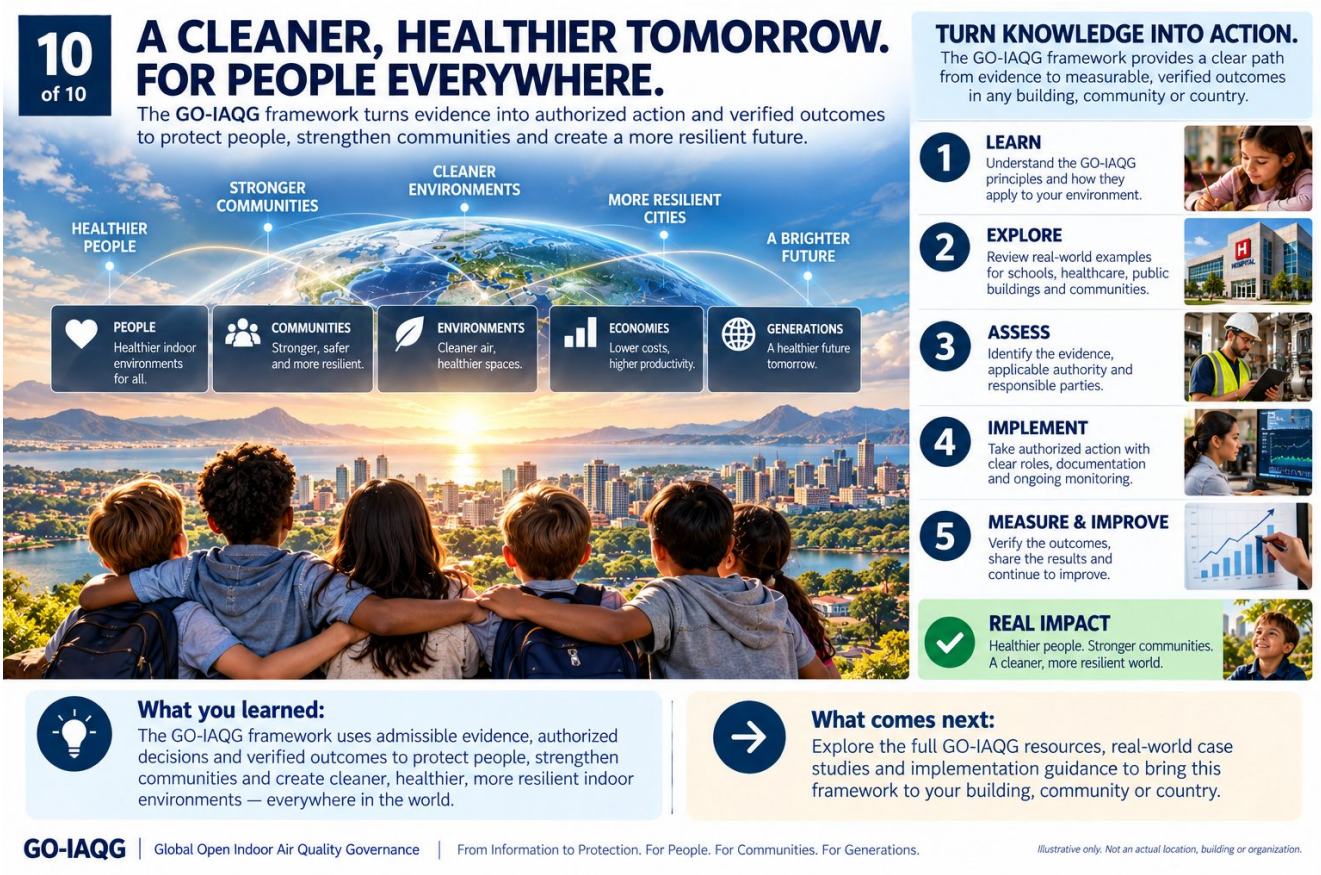
Examination question

Can affected families and communities see the evidence and understand its limits without surrendering privacy?

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10 / 10 A practical pathway for global adoption

GO-IAQG is proposed as a transparent, locally adaptable governance approach, not a substitute for regulators, building codes or public-health practice.



What the image teaches

Begin with a bounded pilot, local authority mapping, minimum evidence specification, independent review plan, explicit stop conditions, outcome metrics and publication of limitations. Cross-border sharing of methods or evidence does not transfer execution authority.

Examination question

Can this community demonstrate a traceable decision and independently reviewable outcome before making broader claims?

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Closing declaration

GO-IAQG is a proposal for connecting environmental knowledge to justified human decisions and independently examinable outcomes. Its value will depend on whether real institutions can test it, identify failures, improve its methods and demonstrate that it makes decisions more accountable without obstructing necessary protective action.

Observed is not inferred. Inferred is not authorized.

A safer building cannot be established by a single attractive number. It requires an appropriate body of evidence, competent human judgment, legitimate authority, documented execution and credible follow-through.

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